

EE344: Milestone 2

MON - 10

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Project Description

Auto-transforming Robot

- To build a multi-modal robot which transforms on user's indication.

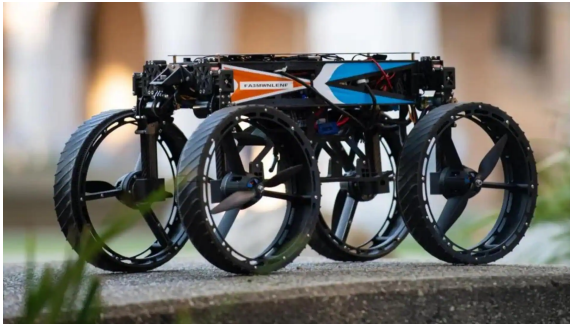


Figure 1: Caltech M4

Milestone 1 Summary

Design Choices:

- Chose SpeedyBee as the main and flight controller.
- Radiomaster for RC control.

Technical Details:

- Weight and power constraints defined (3 kg max, 4S battery).
- BLDC motors selected for drone mode (750 gf thrust per motor).
- High-torque servos for morphing (35 kgcm torque).

Mechanical Structure:

- Chassis dimensions (planned): 28x8 (ground mode), 28x28 (drone mode)
- Material considerations: lightweight, durable (e.g. carbon fiber)

Risks and Mitigation:

- Addressed weight, morphing mechanism, control complexity, rigidity, and high current flow.

Feedback Received

Milestone I review:

- Decide on the material to be used for chassis manufacturing.
- Get an overall cost estimate, accounting for mechanical components.

Problems analysed by us:

- M4 mechanical assembly exceeds weight constraints.
- Chosen IC had inadequate parameters based on simulations

Addressing the issues:

- The material will be decided post-milestone II review.
- Our budget has been capped at INR 70,000.
- Modified the tyre assembly to satisfy weight constraints.
- Verified calculations and found suitable replacement IC.

Project Management Plan and Documentation

Progress

- All tasks allocated in milestone II have been completed.
- PCB has been sent to fabrication after verifying.
- Further work has been started in terms of mechanical design for next milestone.

Management Plan

- Group members have been divided into various subsystems.
- Each group member is responsible for a specific task.
- Flexible task scheduling among subsystem members.
- Critical tasks performed by inter-subsystem groups.

All progress has been documented in OneNote with separate sections for different subsystems.

Gantt Chart Updates

Schematics & Preliminary Analysis				
Choose Flight Controller Firmware	Neel- Sthiti	100%	2/7/25	2/9/25
Go through firmware documentation	Sthiti - Neel	100%	2/10/25	2/15/25
Identify feasible latching mechanisms	Saumya- Neel	100%	2/7/25	2/10/25
Initial Chassis design	Arin - Saumya - Harshil	100%	2/7/25	2/14/25
Initial Tire Assembly Design	Saumya- Arin	100%	2/7/25	2/19/25
Develop Power Distribution Board	Harshil	100%	2/7/25	2/14/25
Initial PCB design	Harshil	100%	2/14/25	2/16/25
Milestone II Evaluation	All	0%	2/20/25	2/28/25

Figure 2: Updated Gantt Chart

Task 1

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Figure 3: Task 1

Choosing Flight Controller Firmware

Available Firmware

- **Betaflight:** Widely used for racing drones and multirotors, known for its performance and customization options.
- **iNav:** Suitable for GPS navigation and waypoint missions.
- **Ardupilot:** A versatile and feature-rich firmware that supports various types of aircraft.

Ardupilot turned out to be the optimal choice, as it is extremely well-documented and is compatible with a wide range of frames and devices. Being one of the most widely accepted open-source firmware for quadrotors, Ardupilot increases our chances of successful execution.

Task 2

Schematics & Preliminary Analysis				
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Figure 4: Task 2

Going through firmware documentation

- Referred and explored the Ardupilot firmware codebase.
- Identified modes of drone operation - Aero, Stabilized, Loiter, Turtle, etc. We will use Stabilized mode for our drone.
- Current plan for integration with drone mode involves creation of new custom modes for ground operation and morphing.
- Successfully tested BLDC motors through Ardupilot using the Mission Planner software (shown in video).
- Simulation for aerial modes explored on Mission Planner.



Task 3

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Figure 5: Task 3

Identify feasible Latching Mechanisms

Mechanism: Push-Push

The latching mechanism locks the arm at a 90° servo rotation. Upon receiving a transform command, the servo briefly pushes the arm up to unlock it, then lowers it to 0° , completing the transition to drive mode.

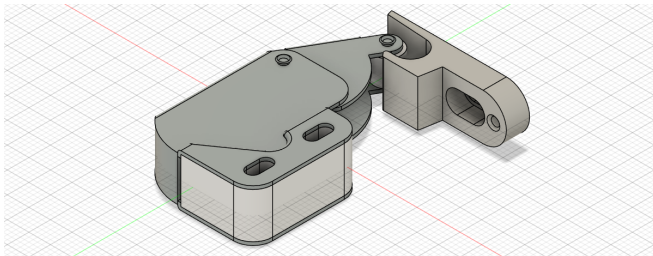


Figure 6: CAD model of a push-push latch

Task 4

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Figure 7: Task 4

Initial Chassis Assembly design

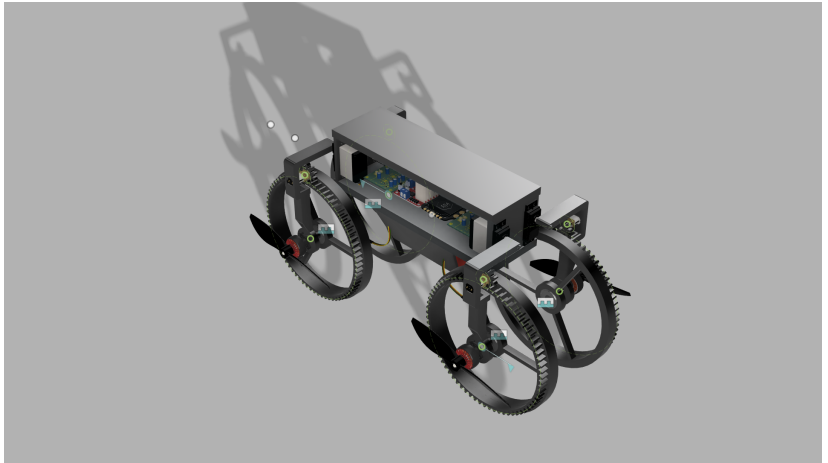


Figure 8: CAD Model of Drive Mode

Initial Chassis Assembly design



Figure 9: CAD Model of Drone Mode

Initial Chassis Assembly design

- The chassis dimensions have been optimized to maintain a square shape in drone mode for effective control.
- A battery compartment has been designed such that the centre of mass of the bot which will majorly be decided by the battery will lie symmetrically to ensure stability.
- All major electrical components have been carefully placed according to their uses.
- The chassis dimensions are 25.07 cm x 8 cm x 8cm.

Task 5

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Figure 10: Task 5

Initial Tire Assembly design

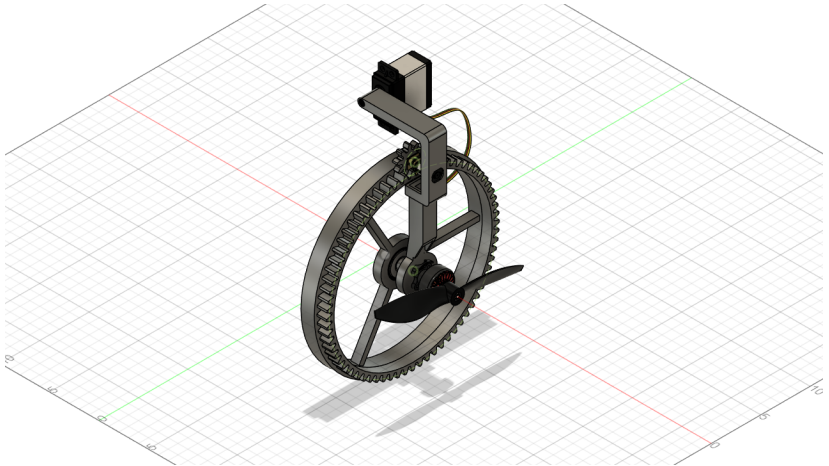


Figure 11: CAD Model of Wheel Assembly

Initial Tire Assembly Design

- Unlike the M4, the propeller and the wheel are placed on opposite sides to reduce the weight cost of bearings.
- The BLDC motor is placed inside the shaft of the wheel, a hollow extension allows for securely connecting the motor to the controller.
- The wheel is actuated using an N20 DC motor through a gear mechanism; our calculations show a max. speed of 12RPM (10cm/sec) in drive mode.
- An L-shaped structure connected to a servo motor acts as our transforming mechanism, allowing for a sleek chassis structure while maintaining a symmetrical drone frame.

Task 6

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Figure 12: Task 6

Power Distribution PCB

- Task 6 and 7 have not only been completed, but exceeded.
- The Power Distribution Board has been designed using the LMR14050 Buck Regulator IC.
- Schematic for one IC and subsequently the overall circuit including 2 Buck IC's, external connections, and test points has been completed.
- PCB layout according to manufacturing constraints has been completed. Board dimensions are 48mm x 40mm.
- The board has been sent for manufacturing to PCB Power.

Technical Design: Buck Schematic

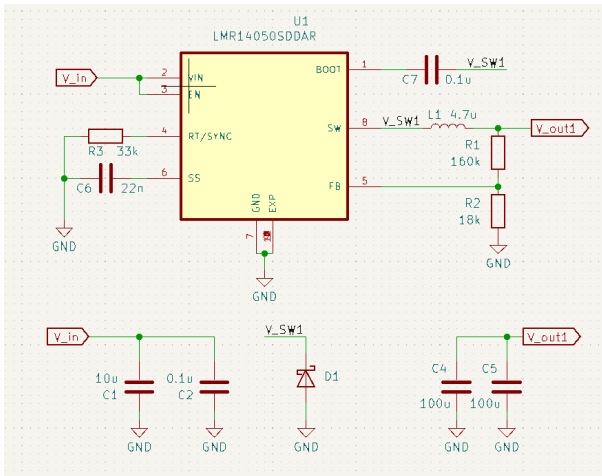


Figure 13: Buck Schematic

Technical Design: Board Layout

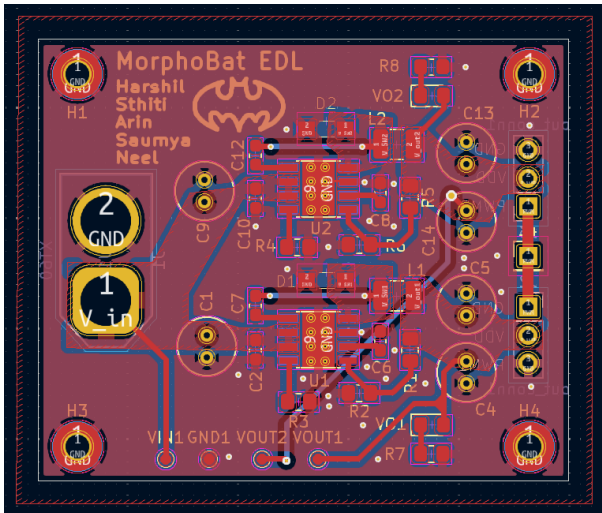


Figure 15: Board Layout

- **Input Voltage:** 14.8V
- **Output Voltage:** 7.4V
- **Maximum Output Current:** 3.8A
- **Transient Response:** 5%
- **Soft-Start Time:** 5ms

$$R_{FBT} = 160k\Omega, \quad R_{FBB} = 18k\Omega$$

Choosing switching frequency $f = 750$ kHz:

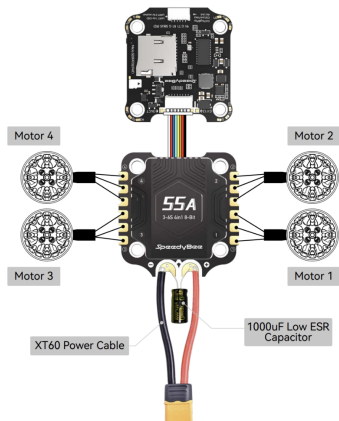
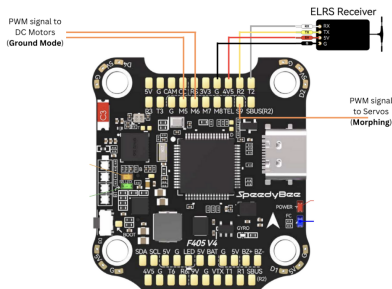
$$L_{\min} = 3.24\mu H$$

Final selection:

$$L = 4.7\mu H \quad (\text{available from WEL})$$

- $R_T = 33k\Omega \Rightarrow f = 732kHz$
- A Schottky diode rated for 5 – 6A is selected.
- Input Capacitor is $10\mu F$ (35V) and $0.1\mu F$ for high-frequency filtering.
- Output Capacitor has two 100uF capacitors.
- Bootstrap capacitor is 0.1uF (given in datasheet)

Circuit Diagram: SpeedyBee Connections



Source: SpeedyBee F405V4 Documentation

Figure 16: Connections to SpeedyBee F405 V4.

Left: Front side, Right: Back side

Block Diagram

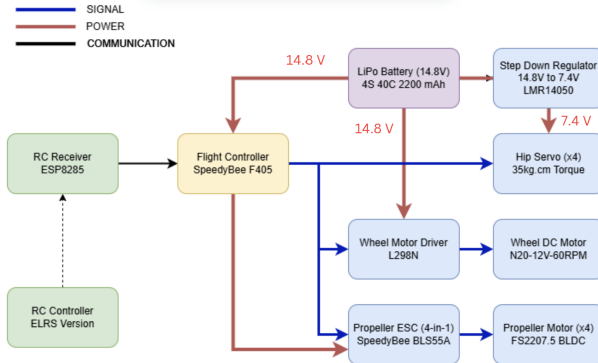


Figure 17: Overall Block Diagram

Testing Plans

Preliminary:

- Testing of BLDC motors using SpeedyBee through Ardupilot

Intermediate:

- Control the 3 sets of motors concurrently using SpeedyBee
- Perform simulations on CAD Assembly model
- Proof-of-Concept testing of initial prototype
- Testing functionality and outputs of PCB

Final:

- Comprehensive testing of deliverables on the final prototype.
- Estimate performance metrics of our final product such as ground speed, morphing time, etc.

Preliminary Testing Results

- Video shows the successful testing of the BLDC motor with SpeedyBee. We have started developing the Drive and Morph modes.
- Once the RC Modules are acquired, we will sync those commands to that of the motors.

Thinking ahead...

- **Software:** Major work will include the creation of custom modes for ground and morphing operations in order to control the required motors as well as the integration of RC modules for manual control.
- **PCB Testing:** A simple PCB testing plan has been created (in OneNote documentation) and will be followed after manufacturing of PCB.
- **Engineering Design:** Moving ahead, we plan to perform extensive simulations on our CAD models while also manufacturing an initial prototype to test for faults and further refine our design.

Thank you!